Problem 1

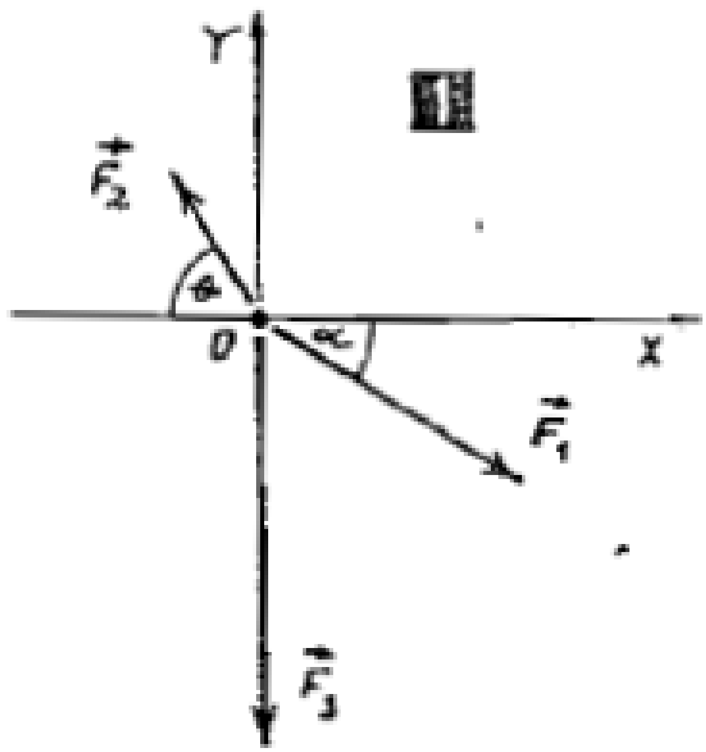

Figure 1: Force directions and angles

Calculate the resultant force $\vec{F}_R$ of three given forces $\vec{F}_1$, $\vec{F}_2$, and $\vec{F}_3$, whose magnitudes are:

$$F_1 = 170 \text{ N}$$
$$F_2 = 100 \text{ N}$$
$$F_3 = 240 \text{ N}$$

The directions and angles are given in the diagram below, where: - $\alpha = 30°$ - $\theta = 60°$

Solution

The components of the resultant force can be found by summing the components of the individual forces along the x- and y-axes:

$$F_{R_x} = \sum F_{x_i} = F_1 \cos(\alpha) - F_2 \cos(\theta)$$
$$F_{R_y} = \sum F_{y_i} = F_2 \sin(\theta) - F_1 \sin(\alpha) - F_3$$

Substitute the given values:

$$
\begin{aligned}
F_{R_x} &= 170\cos(30^\circ) - 100\cos(60^\circ) \\
&= 147.2\ \text{N} - 50\ \text{N} \\
&= 97.2\ \text{N}
\end{aligned}
$$

$$
\begin{aligned}
F_{R_y} &= 100\sin(60^\circ) - 170\sin(30^\circ) - 240 \\
&= 86.6\ \text{N} - 85\ \text{N} - 240 \\
&= -238.4\ \text{N}
\end{aligned}
$$

The magnitude of the resultant force is:

$$
\begin{aligned}
F_R &= \sqrt{F_{R_x}^2 + F_{R_y}^2} \\
&= \sqrt{(97.2)^2 + (-238.4)^2} \\
&= \sqrt{9447.8 + 56830.6} \\
&= \sqrt{66278.4} \\
&= 257.5\ \text{N}
\end{aligned}
$$

Thus, the resultant force is approximately:

$$F_R \approx 257.5\ \text{N}$$

The given solution is correct!